Growth rates compared

Begin with the question: what changes when the input gets bigger?

1. Linear — add the same amount

Linear growth adds the same amount for every equal step in the input.

Example: An odometer records 10 more miles each hour while a vehicle holds 10 miles per hour.

Driving time (hours)	Trip distance (miles)
1	10
2	20
3	30
4	40
5	50
6	60
7	70
8	80

2. Quadratic — the square pattern

Quadratic growth follows the area of a square as its side gets longer.

Example: A square garden bed 2 feet on each side covers 4 square feet; a square foot is a 1-foot square.

Garden bed’s side (feet)	Garden area (square feet)
1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64

“Input” is the number we start with; “output” is what the rule gives back. These tables use different units, so compare their patterns—not miles directly with square feet.

Growth rates compared

Two more patterns: grow in three directions, or keep doubling.

3. Cubic — the cube pattern

Cubic growth follows the volume of a cube as its side gets longer.

Example: A concrete cube 2 feet on each side has 8 cubic feet of volume; a cubic foot is a 1-foot cube.

Cube’s side (feet)	Cube’s volume (cubic feet)
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512

4. Exponential — multiply each time

Exponential growth multiplies by the same factor at each equal step.

Example: Imagine bacteria doubling each hour, starting with one at hour zero.

Time elapsed (hours)	Bacteria in the ideal model
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256

Bacteria cannot double forever: this model assumes unlimited resources. Doubling may start below a square pattern before eventually passing it.

Growth rates compared

One pattern grows especially fast; the other compresses large numbers.

5. Factorial — count possible orders

Factorial growth counts the different ways to place distinct people in a row of seats.

Example: Three different people can occupy a row of three seats in six different ways.

n! (“n factorial”) multiplies whole numbers 1 through group size n; ! is not excitement.

People to seat (n)	Seating arrangements (factorial)
1	1
2	2
3	6
4	24
5	120
6	720
7	5,040
8	40,320

6. Logarithmic — count multiplication steps

Logarithmic growth adds a fixed amount whenever the input grows by a fixed factor.

Example: log10 counts tenfold steps from 1: each step multiplies by ten.

Inputs are powers of ten, not 1–8; “digits” counts written figures (100 has three).

Input number	Tenfold steps (log10)	Written digits
1	0	1
10	1	2
100	2	3
1,000	3	4
10,000	4	5
100,000	5	6
1,000,000	6	7
10,000,000	7	8

Digit counts change in whole-number steps, not along a smooth log curve. “Log10 plus one” gives the digit counts shown **for these powers of ten**.

For these fixed rules, factorial eventually grows fastest; small-input rankings differ. Compare patterns, not quantities with different units.